

Title

Trader Behavior Analysis Based on Bitcoin Fear & Greed Index

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1. Objective

The objective of this project is to analyze the relationship between Bitcoin market sentiment and trader performance using Hyperliquid historical trading data and the Bitcoin Fear & Greed Index.

2. Datasets Used

Historical Trader Data

Contains:

- Account
- Coin
- Side
- Execution Price
- Closed PnL
- Timestamp

Fear & Greed Dataset

Contains:

- Date
- Sentiment Classification
- Fear & Greed Value

3. Methodology

1. Loaded both datasets using Pandas.
2. Converted timestamps into a common date format.
3. Merged datasets on Date.
4. Performed sentiment-wise analysis.
5. Calculated:
 - a. Average PnL
 - b. Total PnL
 - c. Win Rate
 - d. Coin-wise Profitability
 - e. Trade Distribution

4. Key Findings

Finding 1: Extreme Greed Generates Highest Average Profit

Sentiment	Average PnL
Extreme Greed	67.89
Fear	54.29
Greed	42.74
Extreme Fear	34.54
Neutral	34.31

Observation:

Extreme Greed produced the highest average trader profit.

Finding 2: Win Rate Improves During Extreme Greed

Sentiment	Win Rate
Extreme Greed	46.49%
Fear	42.08%
Neutral	39.70%
Greed	38.48%
Extreme Fear	37.06%

Observation:

Winning trades were most common during Extreme Greed.

Finding 3: Most Profitable Coins

Coin	Total Profit
@107	2.78 Million
HYPE	1.95 Million
SOL	1.64 Million
ETH	1.32 Million
BTC	0.87 Million

Observation:

@107, HYPE and SOL generated the largest cumulative profits.

Finding 4: Trade Activity by Market Sentiment

Sentiment	Number of Trades
Fear	61,837
Greed	50,303
Extreme Greed	39,992
Neutral	37,686
Extreme Fear	21,400

Observation:

Most trading activity occurred during Fear and Greed periods.

5. Conclusion

The analysis demonstrates a clear relationship between market sentiment and trading performance. Traders achieved higher profitability and win rates during periods of Extreme Greed, while performance was weaker during Extreme Fear. These findings suggest that market sentiment can serve as a useful indicator for risk management and trading strategy development.